

Design And Analysis of the Sparse Array for DoA Estimation of Noncircular Signals

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Abstract—Recently, many sparse arrays have been proposed to increase the number of degrees of freedom for direction of arrival (DoA) estimation especially for circular signals. Though many practical signals are noncircular, still the properties of noncircularity are hardly exploited in the design of sparse linear arrays. A new array geometry for noncircular signals, which significantly increases the aperture of the virtual array, has been proposed in this paper. The structure is obtained by systematically placing the two uniform linear array (ULA) side by side. The proposed design provides the largest aperture virtual ULA (VULA) as compared to the existing array structures. The aperture of VULA and corresponding weight function are evaluated analytically. The closed-form expression of the Cramér–Rao bound, the minimum lower bound on the error in the DoA estimation, is also obtained for the proposed structure. The performance of the proposed array is studied by the exhaustive computer simulations.

Index Terms—DoA estimation, noncircular signals, nonuniform linear array, virtual array, Cramér Rao bound, weight function, underdetermined system.

I. INTRODUCTION

DIRECTION of Arrival (DoA) estimation is a classical problem in the area of array signal processing. It has a wide range of applications in the field of communications, radar, radio astronomy, sonar, navigation, objects tracking, rescue and emergency assistance, etc. [1]–[8]. Traditional DoA estimation algorithms such as Multiple Signal Classification (MuSiC) [3], ESTimation of Parameters by Rotational Invariant Techniques (ESPRIT) [4], Maximum Likelihood (ML) [5]–[6] etc. are able to resolve only $N - 1$ sources using N sensors [1]–[8].

Many efforts have been made by researchers to get the solution of the underdetermined system, i.e., to resolve more sources using fewer sensors [9]–[27]. They solve this problem using specially designed Non-Uniform Linear Arrays (NULAs) [11]–[27], [36] that provide a larger aperture virtual array/co-array. The virtual sensors in the array come about as a result of the spatial covariance between the sensor output of the array. Minimum Redundancy Array (MRA) [9] and Minimum Hole Array (MHA) [10] are among the first NULA structures

designed for the underdetermined systems. There is no closed form expression for these array geometries and the achievable virtual array aperture, so physical array structures are designed by exhaustive computer search.

Recently, other sparse array configurations have also been proposed, which can be designed and analyzed systematically. One of them is Nested Arrays (NA), which are constructed by concatenating two Uniform Linear Arrays (ULAs) [11]. It can resolve $O(N^2)$ sources using N physical sensors. Co-prime Arrays (CPA) [12] are another NULA structures, which have been designed by interlacing two ULAs of M, N unit inter-sensor spacing. It can resolve $O(MN)$ sources using $M + N - 1$ physical sensors, here M and N are co-prime integers. Others sparse structures such as Coprime Array with Displaced Subarrays (CADiS) [13], Three-Level Prime Array (3LPA) [14], etc. have also been devised to resolve more targets using fewer sensors.

Most of these structures have been designed for circular signal and they use only the spatial covariance lags. But practically many signals are noncircular in nature such as Binary Phase Shift Keying (BPSK), Minimum Shift Keying (MSK), Pulse Amplitude Modulation (PAM), and Unbalance Quadrature PSK (UQPSK), etc. [15]–[24]. Many DoA estimation methods have been proposed for noncircular signals [16]–[18]. These methods achieve better performance in terms of resolution and robustness against noise [16]. But the properties of noncircular signals are hardly exploited, especially in the design of sparse arrays.

The noncircular signals are characterized by non-zero pseudo covariances, which can also be used to enhance the aperture of the virtual array and also improve the performance of any parameter estimation problem by imparting more information. The virtual array obtained using spatial covariances along with pseudo spatial covariances lags is termed as Sum Difference Co-Array (SDCA) because the location of the virtual sensors is obtained by the addition and the subtraction of the physical sensors location.

Here, in this paper the SDCA based virtual array has been driven for noncircular signals but in literature, the concept of SDCA is independently discussed by many authors [25]–[28] without justification. The conventional CPA [25] has been extended with SDCA concept to provide the $MN + 2M + 2N - 1$ virtual sensors using $M + N - 1$ physical sensors. Further, an Adjacent Coprime Array (ADCA) has been presented in [27], which can estimate $N^2 + 3N$ real-valued sources using $2N$ physical sensors. A hand-crafted NULA structure named as Sum-Difference Composite Co-array (SDCC) with 6 physical sensors has been presented in [26] and has the highest virtual

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array aperture among all 6 sensor NULAs. But the SDCC [26] design is limited to 6 sensors and there is no general design methodology for designing similar arrays for other than 6 sensors [26].

In this paper, we present a complete methodology for designing a generic NULA, such that the corresponding Virtual Uniform Linear Array (VULA) aperture is highest in comparison to earlier suggested arrays [25]–[27]. The proposed array has been designed by placing the two ULAs side by side. The inter-sensor spacing for both the ULAs and the spacing between two ULAs have been evaluated such that the designed VULA aperture is maximum for the given number of physical sensors. The length of both ULAs, i.e., the distribution of total physical sensors between both the ULAs is also optimized to maximize the aperture of the VULA. We have also derived the expression for the aperture of designed VULA as a function of total number of physical sensors.

A vital performance metric of these virtual arrays is the weight function, which is used to study their stability. In this paper, we have derived the weight function for the proposed design. Moreover, the Cram r Rao bound (CRB), a universal tool for evaluating the performance of any parameter estimation algorithm, is derived for DoA estimation using the proposed structure.

The designed array structure provides the maximum aperture VULA for the given number of physical sensors in most of the cases except only the 6 physical sensor structure. In this case proposed VULA aperture is equal to the VULA aperture [26] of SDCC structure.

The performance of various DoA estimation algorithms for the designed structure is studied and compared with the other similar arrays. Numeric simulations show the efficacy of the proposed structure over earlier suggested NULAs [25]–[27]. Apart from providing the larger aperture VULA, the proposed array also increases the estimation accuracy. The weight function, CRB and numeric simulation analysis show that the proposed design is more stable and more accurate in estimating the DoA as compared to other existing structures [25]–[26].

II. SIGNAL MODEL

Let M narrow band, uncorrelated, noncircular and far field sources impinge on an array $\mathbb{S}_{pa} = \{x_1, x_2, \dots, x_N\}$ with N sensors from the direction $\{\theta_1, \dots, \theta_M\}$ w.r.t. the normal of the array. Mathematically, the array output at the n th sensor corresponding to the k th snapshot can be expressed as

$$y_n(k) = \sum_{l=1}^M a_n(\theta_l) s_l(k) + v_n(k) \quad (1)$$

$1 \leq k \leq K \quad 1 \leq n \leq N$

where

$$a_n(\theta_l) = e^{j\pi x_n \sin(\theta_l)} \quad (2)$$

x_n denotes the location of the n th sensor expressed in terms of $\lambda/2$ (one unit separation = $\lambda/2$), λ is the wavelength of the received signal, K is the total number of snapshots, $s_l(k) \in \mathbb{C}$ denotes the signal of the l th source corresponding to the k th

snapshot. $v_n(k)$ denotes noise at the n th sensor corresponding to the k th snapshot, which is uncorrelated with all the sources.

Let $\mathbf{y}(k) = [y_1(k), \dots, y_N(k)]^T$, where superscript $(\cdot)^T$ denotes matrix transpose operation, be the $N \times 1$ array observation vector. Using $\mathbf{a}(\theta_l) = [a_1(\theta_l), a_2(\theta_l), \dots, a_N(\theta_l)]^T$, $\mathbf{y}(k)$ can be written as,

$$\mathbf{y}(k) = \sum_{l=1}^M \mathbf{a}(\theta_l) s_l(k) + \mathbf{v}(k) = \mathbf{A}\mathbf{s}(k) + \mathbf{v}(k) \in \mathbb{C}^{N \times 1} \quad (3)$$

where $\mathbf{A} = [\mathbf{a}(\theta_1), \mathbf{a}(\theta_2), \dots, \mathbf{a}(\theta_M)]$ is the $N \times M$ array manifold matrix, $\mathbf{s}(k) = [s_1(k), \dots, s_M(k)]^T$ is the $M \times 1$ source signal vector and $\mathbf{v}(k) = [v_1(k), v_2(k), \dots, v_N(k)]^T$ is the $N \times 1$ noise vector corresponding to the k th snapshot.

The covariance matrix $\mathbf{R}_y \in \mathbb{C}^{N \times N}$ of the received signal $\mathbf{y}(k)$ is given as,

$$\mathbf{R}_y = E[\mathbf{y}(k) \mathbf{y}^H(k)] = \sum_{l=1}^M g_l \mathbf{a}(\theta_l) \mathbf{a}^H(\theta_l) + \sigma_n^2 \mathbf{I} \quad (4)$$

where $(\cdot)^H$ denotes the matrix Hermitian operation and $g_l = E[s_l(k) s_l^H(k)]$. $\mathbf{R}_y$ is complex, Hermitian and positive definite matrix. The pseudo covariance matrix $\mathbf{\Gamma}_y \in \mathbb{C}^{N \times N}$ of $\mathbf{y}(k)$ is defined as,

$$\mathbf{\Gamma}_y = E[\mathbf{y}(k) \mathbf{y}^T(k)] = \sum_{l=1}^M \tilde{g}_l \mathbf{a}(\theta_l) \mathbf{a}^T(\theta_l) \quad (5)$$

with $\tilde{g}_l = E[s_l(k) s_l^T(k)]$. $\mathbf{\Gamma}_y$ is a complex symmetric pseudo covariance matrix which characterizes the noncircular signals. The value of $\mathbf{\Gamma}_y$ is zero for circular signals. Noncircular signals provide an additional independent information in terms of $\mathbf{\Gamma}_y$ which can be exploited to provide significant performance gain in any parameter estimation problem.

In order to take advantage of the benefits associated with noncircular source signals, we define the extended observation vector $\mathbf{z}(k) \in \mathbb{C}^{2N \times 1}$ as,

$$\mathbf{z}(k) = [\mathbf{y}^T(k), \mathbf{y}^H(k)]^T$$

The covariance matrix $\mathbf{R}_z \in \mathbb{C}^{2N \times 2N}$ corresponding to the extended observation vector $\mathbf{z}(k)$ is given as,

$$\mathbf{R}_z = E[\mathbf{z}(k) \mathbf{z}^H(k)] = \begin{bmatrix} \mathbf{R}_y & \mathbf{\Gamma}_y \\ \mathbf{\Gamma}_y^* & \mathbf{R}_y^* \end{bmatrix} \quad (6)$$

where superscript $(\cdot)^*$ denotes matrix conjugate operation. Here, the covariance matrix $\mathbf{R}_z$ of the extended observation vector is used to construct an extended aperture virtual array. The $2N \times 2N$ dimensional covariance matrix $\mathbf{R}_z$ is the positive definite and Hermitian. A $N \times N$ Hermitian matrix has $(N^2 + N)/2$ unique entries [29]. Therefore, the covariance matrix $\mathbf{R}_z$ will have $2N^2 + N$ unique entries which can be used to extend the virtual array aperture.

By vectorizing $\mathbf{R}_z$, a new vector $\mathbf{r}_z$ with $4N^2 \times 1$ dimension is obtained as,

$$\mathbf{r}_z = \text{vec}(\mathbf{R}_z) \quad (7)$$

where $\text{vec}(\cdot)$ is the “vectorization” operator that turns a matrix into a vector by stacking the columns of the matrix one below to another [31].

Mathematically, the vectorized version of covariance matrix $\mathbf{R}_z$ is simplified in the following Lemma 1.

Lemma 1: The vectorization of covariance matrix $\mathbf{R}_z$ can be expressed as,

$$\mathbf{r}_z = \mathbf{J}\mathbf{B}\mathbf{g}_s + \mathbf{v} \quad (8)$$

where $\mathbf{B}$ is given in (12), $\mathbf{g}_s = [\mathbf{g}^T \ \tilde{\mathbf{g}}^H \ \tilde{\mathbf{g}}^T \ \mathbf{g}^H]^T \in \mathbb{C}^{4M \times 1}$ with $\mathbf{g} = [g_1, \dots, g_M]^T$ and $\tilde{\mathbf{g}} = [\tilde{g}_1, \dots, \tilde{g}_M]^T$ denotes the vector of covariance and pseudo covariance of M sources and $\mathbf{J} \in \mathbb{R}^{4N^2 \times 4N^2}$ is the permutation matrix defined in (10).

Proof: As given in (6) the covariance matrix $\mathbf{R}_z$ is a block matrix. The relation between the block matrix vectorization $\text{vec}_b(\cdot)$ and vectorization of a matrix, $\text{vec}(\cdot)$ is [30] given as,

$$\begin{aligned} \mathbf{r}_z &= \text{vec}(\mathbf{R}_z) = \mathbf{J}\text{vec}_b(\mathbf{R}_z) \\ &= \mathbf{J}[\text{vec}(\mathbf{R}_y) \ \text{vec}(\mathbf{\Gamma}_y^*) \ \text{vec}(\mathbf{\Gamma}_y) \ \text{vec}(\mathbf{R}_y^*)]^T \end{aligned} \quad (9)$$

where $\mathbf{J} \in \mathbb{R}^{4N^2 \times 4N^2}$ denotes the permutation matrix [30], defined as follow,

$$\mathbf{J} = \left([\mathbf{I}]_{2 \times 2} \otimes [\mathbf{P}]_{2N \times 2N} \otimes [\mathbf{I}]_{N \times N} \right)^{-1} \quad (10)$$

Here $\mathbf{P} = \sum_{i=1}^N \sum_{j=1}^2 \mathbf{H}_{ij} \otimes \mathbf{H}_{ij}^T$ and $\mathbf{H}_{ij} \in \mathbb{C}^{2N \times 2N}$ having unity element in (i, j) th position and zeros elsewhere. From (4), (5) and using the identity $\text{vec}(\mathbf{a}\mathbf{b}^H) = \mathbf{b}^* \otimes \mathbf{a}$ where $\mathbf{a}, \mathbf{b} \in \mathbb{C}^{M \times 1}$ [31], we can simplify (9) as,

$$\mathbf{r}_z = \mathbf{J} \begin{bmatrix} \sum_{i=1}^M (\mathbf{a}^*(\theta_i) \otimes \mathbf{a}(\theta_i))g_i + \sigma_n^2 \mathbf{1}_n \\ \sum_{i=1}^M (\mathbf{a}^*(\theta_i) \otimes \mathbf{a}^*(\theta_i))\tilde{g}_i^* \\ \sum_{i=1}^M (\mathbf{a}(\theta_i) \otimes \mathbf{a}(\theta_i))\tilde{g}_i \\ \sum_{i=1}^M (\mathbf{a}(\theta_i) \otimes \mathbf{a}^*(\theta_i))\tilde{g}_i^* + \sigma_n^2 \mathbf{1}_n \end{bmatrix} \quad (11)$$

where the symbol $\otimes$ denotes the left Kronecker product [1] and $\mathbf{1}_n = \text{vec}(\mathbf{I}) = [\mathbf{e}_1^T \dots \mathbf{e}_N^T]^T$ with $\mathbf{e}_i \in \mathbb{R}^{N \times 1}$ being a column vector of all zeros except a 1 at the i th position. Define,

$$\begin{aligned} \mathbf{B}_{\mathbb{D}}^+ &= [\mathbf{a}^*(\theta_1) \otimes \mathbf{a}(\theta_1) \dots \mathbf{a}^*(\theta_M) \otimes \mathbf{a}(\theta_M)] \\ \mathbf{B}_{\mathbb{S}}^+ &= [\mathbf{a}(\theta_1) \otimes \mathbf{a}(\theta_1) \dots \mathbf{a}(\theta_M) \otimes \mathbf{a}(\theta_M)] \\ \mathbf{B}_{\mathbb{S}}^- &= [\mathbf{a}^*(\theta_1) \otimes \mathbf{a}^*(\theta_1) \dots \mathbf{a}^*(\theta_M) \otimes \mathbf{a}^*(\theta_M)] \\ \mathbf{B}_{\mathbb{D}}^- &= [\mathbf{a}(\theta_1) \otimes \mathbf{a}^*(\theta_1) \dots \mathbf{a}(\theta_M) \otimes \mathbf{a}^*(\theta_M)] \end{aligned}$$

Using the above definition in (11), we get

$$\mathbf{r}_z = \mathbf{J}\mathbf{B}\mathbf{g}_s + \mathbf{v} \quad (12)$$

where

$$\mathbf{B} = \begin{bmatrix} \mathbf{B}_{\mathbb{D}}^+ & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{B}_{\mathbb{S}}^- & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{B}_{\mathbb{S}}^+ & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{B}_{\mathbb{D}}^- \end{bmatrix}$$

where $\mathbf{0}$ is the zero matrix of the size $N^2 \times M$, $\mathbf{g}_s = [\mathbf{g}^T \ \tilde{\mathbf{g}}^H \ \tilde{\mathbf{g}}^T \ \mathbf{g}^H]^T$ and $\mathbf{v} = \mathbf{J}[\sigma_n^2 \mathbf{1}_n \ \mathbf{0} \ \mathbf{0} \ \sigma_n^2 \mathbf{1}_n]^T$. ■

The vectorized covariance matrix is the main function of $\mathbf{B} \in \mathbb{C}^{4N^2 \times 4M}$ matrix and $4M \times 1$ dimensional $\mathbf{g}_s$ vector. To get more insight into it, let us analyze vector $\mathbf{r}_z$. As $\mathbf{J}$ matrix is just a permutation matrix, to understand the physical contribution corresponding to each entry of $\mathbf{r}_z$, it is analyzed without the $\mathbf{J}$ matrix. First N^2 entries of $\mathbf{r}_z$ in no noise case are given by the product of matrix $\mathbf{B}_{\mathbb{D}}^+$ and vector $\mathbf{g}$, i.e.,

$$[\mathbf{r}_z]_{1:N^2} \approx \mathbf{B}_{\mathbb{D}}^+ \mathbf{g}$$

$\mathbf{b}_{\mathbb{D}_i}^+$, the i th column of $\mathbf{B}_{\mathbb{D}}^+$, is given as,

$$\mathbf{b}_{\mathbb{D}_i}^+ = \mathbf{a}^*(\theta_i) \otimes \mathbf{a}(\theta_i) \quad (13)$$

and the corresponding t th elements of $\mathbf{b}_{\mathbb{D}_i}^+$ is written as,

$$\begin{aligned} \mathbf{b}_{\mathbb{D}_i}^+(t) &= e^{-j\pi x_p \sin(\theta_i) + j\pi x_q \sin(\theta_i)} = e^{j\pi(-x_p + x_q) \sin(\theta_i)} \\ 0 \leq p, q \leq N-1 \quad t &= (p-1) + (N-1)(q-1) \end{aligned} \quad (14)$$

where x_p, x_q denotes the position of p th and q th physical sensor respectively in terms of $\lambda/2$, (i.e., one unit spacing). As term (2), $e^{j\pi x \sin(\theta_i)}$ represents the steering gain of the sensor located at x for the signal coming from direction θ_i . Therefore, $\mathbf{b}_{\mathbb{D}_i}^+(t)$ corresponds to the steering gain corresponding to the sensor present at $x_q - x_p$ position to the signal coming from θ_i direction. Therefore, the vector $\mathbf{b}_{\mathbb{D}_i}^+$ can be interpreted as the steering vector for the signal coming from θ_i direction for the co-array $\mathbf{S}_{\mathbb{D}}^+$, given as,

$$\mathbf{S}_{\mathbb{D}}^+ = \{x_q - x_p\}, \quad \forall p, q = 1, \dots, N \quad (15)$$

Hence, first N^2 entries of $\mathbf{r}_z$ can be seen as one snapshot of the signal $\{g_1 \dots g_M\}$ received at an array whose sensors are located at $\mathbf{S}_{\mathbb{D}}^+$ from directions $\{\theta_1 \dots \theta_M\}$. Here, $\mathbf{S}_{\mathbb{D}}^+$ is termed as positive difference co-array. The next entries of vector $\mathbf{r}_z$ are given as,

$$[\mathbf{r}_z]_{N^2+1:2N^2} \approx \mathbf{B}_{\mathbb{S}}^- \tilde{\mathbf{g}}^*$$

The i th column of $\mathbf{B}_{\mathbb{S}}^-$ is given as,

$$\mathbf{b}_{\mathbb{S}_i}^- = \mathbf{a}^*(\theta_i) \otimes \mathbf{a}^*(\theta_i) \quad (16)$$

and t th elements of $\mathbf{b}_{\mathbb{S}_i}^-$ is written as,

$$\begin{aligned} \mathbf{b}_{\mathbb{S}_i}^-(t) &= e^{-j\pi x_p \sin(\theta_i) - j\pi x_q \sin(\theta_i)} = e^{j\pi(-x_p - x_q) \sin(\theta_i)} \\ 0 \leq p, q \leq N-1 \quad t &= (p-1) + (N-1)(q-1) \end{aligned} \quad (17)$$

Following the above discussion, $\mathbf{b}_{\mathbb{S}_i}^-$ can be interpreted as the steering vector corresponding to the signals coming from θ_i direction and the array whose sensors positions are given by $\mathbf{S}_{\mathbb{S}}^-$,

$$\mathbf{S}_{\mathbb{S}}^- = \{-x_p - x_q\}, \quad \forall p, q = 1, \dots, N \quad (18)$$

The array $\mathbf{S}_{\mathbb{S}}^-$ is termed as the negative sum co-array. $\mathbf{B}_{\mathbb{S}}^+$ and $\mathbf{B}_{\mathbb{D}}^-$ are interpreted as the steering matrix corresponding to positive sum co-array ($\mathbf{S}_{\mathbb{S}}^+$) and negative difference co-array

$(\mathbf{S}_{\mathbb{D}}^-)$ respectively given as,

$$\mathbf{S}_{\mathbb{S}}^+ = \{x_p + x_q\}, \quad \forall p, q = 1, \dots, N \quad (19)$$

$$\mathbf{S}_{\mathbb{D}}^- = \{-x_p + x_q\}, \quad \forall p, q = 1, \dots, N \quad (20)$$

Therefore, the complete vector $\mathbf{r}_z$ can be seen as the signal received from the $\{\theta_1 \dots \theta_M\}$ directions by the virtual array $\mathbb{S}$ which is the union of all defined co-arrays in (15), (18)–(20), i.e.,

$$\mathbb{S} = \mathbf{S}_{\mathbb{S}}^+ \cup \mathbf{S}_{\mathbb{S}}^- \cup \mathbf{S}_{\mathbb{D}}^+ \cup \mathbf{S}_{\mathbb{D}}^- \quad (21)$$

This complete virtual array $\mathbb{S}$ is termed as Sum-Difference Co-Array (SDCA), which might contain sensors from $\{-2x_N, \dots, 2x_N\}$ where x_N denotes the location of the N th sensor in an array but all the sensors may not be continuous. The continuous subset of this complete virtual array $\mathbb{S}$ is termed as the Virtual Uniform Linear Array (VULA).

Here, we have modeled $\mathbf{r}_z$ as the received signal from the M directions by an array $\mathbb{S}$. It should be noted that the signal value corresponding to the sensor of co-arrays $\mathbf{S}_{\mathbb{D}}^+$ and $\mathbf{S}_{\mathbb{D}}^-$ are $\{g_1 \dots g_M\}$ and $\{g_1^* \dots g_M^*\}$ respectively and at $\mathbf{S}_{\mathbb{S}}^+$ and $\mathbf{S}_{\mathbb{S}}^-$ are $\{\tilde{g}_1 \dots \tilde{g}_M\}$ and $\{\tilde{g}_1^* \dots \tilde{g}_M^*\}$ respectively. As we are dealing with noncircular signals ($\tilde{g} \neq 0$), hence we have additional virtual sensors corresponding to $\mathbf{S}_{\mathbb{S}}^+$ and $\mathbf{S}_{\mathbb{S}}^-$.

The aperture of virtual array $\mathbb{S}$ increases due to the utilization of pseudo covariance matrix. It should also be pointed here that the pseudo covariance matrix bring more information which also enhances the accuracy of DoA estimation.

Consequently, $\mathbf{r}_z$ alias a single snapshot of received vector by a virtual array $\mathbb{S}$. The vector $\mathbf{r}_z$ can be used to solve the underdetermined system using various algorithms [11]–[13], [32]–[35]. The details of each of these algorithms are well documented in the literature [11], [13], [32]–[35] but for the sake of completeness, we have described one algorithm in Appendix-I.

Most of the algorithms [11], [13], [32]–[35] use only the VULA to estimate DoA. Therefore, the requirement is to design a physical array such that the corresponding VULA has maximum aperture.

III. PROPOSED ARRAY CONFIGURATION

The total number of degrees of freedom available directly depends upon the VULA aperture. Here, we propose a methodology to design a sparse array which has the largest aperture VULA. The proposed array, termed as the Nested Array with Displaced Subarray (NADiS), is designed by placing the two ULAs side by side. First ULA ($\mathbb{S}_1$) has N_1 sensors with one unit inter-sensor spacing, i.e., $\lambda/2$ and the second ULA ($\mathbb{S}_2$) has N_2 sensors and it is placed at a distance of L units from the first sensor of ULA-I. ULA-II has inter-sensor spacing of D units. The complete design of the proposed array is as shown in Fig. 1. Mathematically,

$$\mathbb{S}_1 = \{l | l = m, 0 \leq m \leq N_1 - 1\} \quad (22)$$

$$\mathbb{S}_2 = \{l | l = L + nD, 0 \leq n \leq N_2 - 1\} \quad (23)$$

Theorem 1 describes the selection rule for the value L and D that maximize the aperture of VULA.

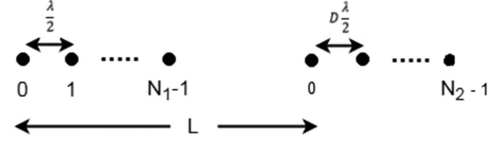


Fig. 1. Proposed Array Configuration.

Theorem 1: Consider a N sensor NADiS structure consisting of two ULAs as ULA-I and ULA-II with $\frac{\lambda}{2}$ and $D\frac{\lambda}{2}$ inter-sensor spacing respectively. Let ULA-I contains N_1 sensors and therefore, ULA-II contain $N - N_1$ sensors. To achieve the maximum continuous lags, ULA-II should be placed at $L = 3N_1 - 2$ unit distance from the first sensor of ULA-I. The value of inter-sensor spacing of ULA-II, D for the largest aperture VULA should be $D = 2N_1 - 1$.

We state the following proposition before proving Theorem 1.

Proposition 1: Let $\mathbb{G}$ be an ULA of N sensors having one unit inter-sensor spacing, i.e., $\lambda/2$. The complete SDCA corresponding to $\mathbb{G}$ is continuous from $\{-2(N-1) \dots 2(N-1)\}$. Hence, the aperture of corresponding VULA is from $-2(N-1)$ to $2(N-1)$.

Proof of Theorem 1: For the ULA-I consisting of N_1 sensors, the aperture of VULA can be obtained using proposition-1, which is from $-2(N_1-1)$ to $2(N_1-1)$. Therefore, the next lag is required at $2(N_1-1) + 1$ so ULA-II should be placed such that,

$$2(N_1 - 1) + 1 = L - (N_1 - 1) \quad (24)$$

Therefore, the value of L should be,

$$L = 3N_1 - 2$$

Next, we consider the selection of parameter D such that the proposed structure does not contain any holes. For the given L , we achieve the continuous lags till $L + (N_1 - 1)$. Therefore, the next lag $L + (N_1 - 1) + 1$ is obtained by placing the next sensor in ULA-II as,

$$L + (N_1 - 1) + 1 = L + D - (N_1 - 1) \quad (25)$$

Therefore, the value of D is,

$$D = 2N_1 - 1 \quad (26)$$

This completes our objective of designing the maximum aperture VULA. ■

The complete physical NADiS array can be designed using the Theorem 1. Further, we can also evaluate the aperture of the corresponding VULA. Based on the value of L and D as given in the Theorem 1, the following Lemma gives the number of consecutive lags in VULA.

Lemma 2: The aperture of VULA corresponding to the proposed NADiS configuration is

$$p(N_1, N_2) = 4N_1(N_2 + 1) - 2N_2 - 3$$

Proof: According to the proposed array structure, given in Fig. 1, the last sensor of ULA-II is placed at

$$L + (N_2 - 1)D$$

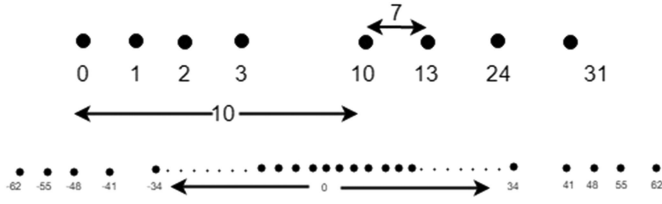


Fig. 2. An example of NADiS configuration where $N = 8$, $N_1 = 4$, $N_2 = 4$ and $L = 10$ $D = 7$. (a) The NADiS configuration. (b) The virtual array.

Hence, the last continuous lag obtained using the last sensor of ULA-II is given as,

$$L + (N_2 - 1)D + N_1 - 1$$

By substituting the value of L and D from Theorem 1, the one side aperture of VULA is,

$$2N_1(N_2 + 1) - N_2 - 2 \quad (27)$$

Hence the two-sided aperture of VULA is given as,

$$\begin{aligned} p(N_1, N_2) &= 2(2N_1(N_2 + 1) - N_2 - 2) + 1 \\ &= 4N_1(N_2 + 1) - 2N_2 - 3 \end{aligned} \quad (28)$$

■

Using the above formulation, we can design a new sparse array for any number of physical sensors, such that the aperture of VULA is maximum.

This design is better illustrated with the help of an example.

Example 1: Consider the NADiS structure with $N = 8$ physical sensors with $N_1 = 4$ and $N_2 = 4$. Hence, the ULA-I has the 4 sensors at $\{0, 1, 2, 3\}$ location and ULA-II has 4 sensors. From Theorem 1, the inter-array spacing $L = 3N_1 - 2$ is 10 units and the inter-sensor spacing for ULA-II is $D = 2N_1 - 1 = 7$ units. So ULA-II sensors are placed at $\{10, 10 + 7 = 17, 10 + 2 * 7 = 24, 10 + 3 * 7 = 31\}$. Hence, the array of 8 physical sensors is $\{0, 1, 2, 3, 13, 22, 31\}$ and its VULA aperture is from $-(2N_1(N_2 + 1) - N_2 - 2) = -34$ to $(2N_1(N_2 + 1) - N_2 - 2) = 34$.

Fig. 2 shows the VULA aperture is -34 to 34 which is equal to the analytically calculated aperture. The VULA aperture can be maximized by optimizing the distribution of the physical sensors.

The proposed NADiS design is based upon the two ULAs, the length of these ULAs can be further optimized to maximize the virtual aperture. Now we address the problem of maximizing the aperture of the VULA under the constraint of a fixed number of physical sensors. The following Theorem provides the optimal sensor distribution among both ULAs, i.e., ULA-I and ULA-II, which maximizes the aperture of VULA.

Theorem 2: The proposed array of N sensors consists of two uniform linear subarrays with N_1 and N_2 sensors. The aperture of VULA is maximum when

$$N_1 = \frac{N}{2} + \frac{3}{4}, \quad N_2 = \frac{N}{2} - \frac{3}{4} \quad (29)$$

Proof: The optimum distribution of total number of available sensors N into two subarrays which increase the aperture

of VULA can be obtained by solving a simple optimization problem as,

$$\begin{aligned} \max \quad & p(N_1, N_2) \\ \text{s.t.} \quad & N_1 + N_2 = N \end{aligned}$$

The aperture of VULA ($p(N_1, N_2)$) for a given values of N_1 and N_2 is given in (28),

$$4N_1(N_2 + 1) - 2N_2 - 3$$

Moreover, $N_2 = N - N_1$, the aperture of VULA can be written as,

$$4N_1(N - N_1 + 1) - 2N + 2N_1 - 3$$

The aperture of VULA is the function of N_1 , so the optimum value of N_1 such that the aperture of VULA is maximum, can be formulated as

$$\begin{aligned} \max_{N_1} \quad & (4N_1(N - N_1 + 1) - 2N + 2N_1 - 3) \\ \text{subject to} \quad & N_1 > 0 \end{aligned} \quad (30)$$

Differentiating the above equation w.r.t. N_1 and equating to zero. We get,

$$\begin{aligned} N_1 &= \frac{N}{2} + \frac{3}{4} \\ N_2 &= \frac{N}{2} - \frac{3}{4} \end{aligned} \quad (31)$$

After substituting the optimum value of N_1 in Theorem 1 and Lemma 2, we get

$$\begin{aligned} L &= \frac{3N}{2} + \frac{1}{4} \\ D &= N + \frac{1}{2} \end{aligned}$$

and the two sided aperture of VULA is,

$$p(N) = N^2 + N - \frac{3}{4} \approx N^2 + N - 1$$

■

In the previous example, we had considered a NADiS of 8 physical sensors, according to Theorem 2 the optimum distribution of N should be $N_1 = \frac{N}{2} + \frac{3}{4} = 4.75 \approx 5$, and $N_2 = \frac{N}{2} - \frac{3}{4} = 3.25 \approx 3$. Accordingly, the previous designed array was not optimized, now we consider an optimized 8 sensors array.

Example 2: An optimum NADiS with 8 physical sensors have 5 sensors at $\{0, 1, 2, 3, 4\}$ location in the ULA-I and 3 sensors in ULA-II for maximum VULA aperture. From Theorem 1 inter-array spacing is $L = 3N_1 - 2 = 13$ units and inter-sensor spacing for ULA-II is $D = 2N_1 + 1 = 9$ units. So ULA-II sensors place at $\{13, 13 + 9 = 22, 13 + 2 * 9 = 31\}$. Hence, the array of 8 physical sensors is $\{0, 1, 2, 3, 13, 22, 31\}$ as shown in the Fig. 3a. Fig. 3b shows the corresponding virtual array. It can be observed that the proposed array provides 71 consecutive lags from -35 to 35 . The optimum aperture of the virtual array is 71 which is higher than the NADiS with the choice of $N_1 = N_2 = 4$.

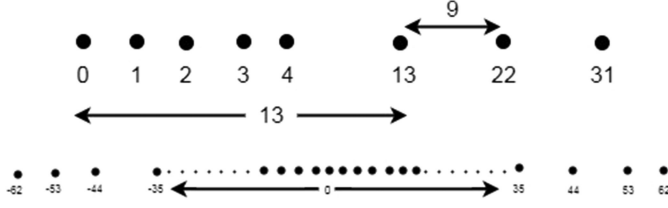


Fig. 3. An example of NADiS configuration where $N = 8$, $N_1 = 5$, $N_2 = 3$ and $L = 13$, $D = 9$. (a) The NADiS configuration. (b) The virtual array.

TABLE I
COMPARISON OF LAGS OBTAINED BY DIFFERENT ARRAY CONFIGURATION

Number of sensors	Different Array Configuration			
	ADCA [27]	CPASD [25]	SDCC [26]	Proposed array
N=6	37	37	41	41
N=8	57	57	NA	71
N=10	81	81	NA	109
N=12	109	109	NA	155

The designed NADiS is optimum in the sense that it uses almost all the unique entries of the $\mathbf{R}_z$ matrix to generate largest aperture VULA. Next, we summarize the array designing steps before proceeding to study the performance analytically.

- For the given number of total physical sensors, first we determine the optimum distribution of the total number of sensors into two subarrays using the equation (29) given in Theorem 2.
- Next step is to calculate inter-sensor spacing D for ULA-II and inter-array distance L between two subarrays using the Theorem 1.
- Substitute the value of L and D in equations (22) and (23) and determine the location of physical sensors.
- The designed array provides the virtual array apertures as given in equation (28).

With the above algorithm, an optimum physical array can be designed such that the corresponding VULA has maximum aperture. The proposed array configuration is optimum only for noncircular signals.

IV. COMPARISON OF DIFFERENT NESTED ARRAY STRUCTURES

In literature, many sparse arrays have been discussed which have various capabilities. Here, we compare the proposed array design with ADCA [27], CPASD [25] and SDCC [26] array structures in terms of the aperture of VULA. These array structures use the SDCA concept to construct virtual arrays. Later, we will also compare them on the basis of source resolving capabilities using numeric simulations.

Table I compares the aperture of VULA for each array structure where the total number of physical sensors are chosen to be 6, 8, 10 and 12. From the Table I, the NADiS structure (with optimal sensor distribution) and SDCC [26] has equal number of continuous lags for 6 physical sensors but SDCC [26] is a handcrafted structure so it can not be used to design any other

dimensional array. Moreover, the proposed structure has maximum virtual aperture for all size arrays. Though, we have given results up to 12 sensors physical array, it can be easily seen that the trend holds for other sizes too.

V. WEIGHT FUNCTION

An important parameter in designing VULA is the weight function. As we discussed above the location of each sensor of virtual array is determined by either the addition or the subtraction of the location of physical sensor. Therefore, virtual sensors might have more than one occurrence as addition and subtraction of different sensor location results in the same value. The output of each virtual sensor is obtained by averaging corresponding occurrences. Definitely, this averaging makes the estimate of the corresponding virtual sensor better by decreasing its variance [36] and makes the output more stable. Formally, the stability of the designed array is studied using the weight function.

The weight function is defined as the frequency of occurrence of each virtual sensor in the complete virtual array. Mathematically, it is defined as,

Definition 1: The weight function $f(n)$, counts the periodicity of each element n of $\mathbb{S}$, defined as

$$\mathbb{F}_n = \left\{ (x_p, x_q) \left| \begin{cases} x_p, x_q \in \mathbb{S}_{pa} \\ x_p - x_q = n \\ x_p + x_q = n \\ -x_p - x_q = n \\ x_q - x_p = n \end{cases} \right. \right\} \quad n \in \mathbb{S}$$

$$f(n) = |\mathbb{F}_n|$$

where $|\mathbb{k}|$ is the cardinality of the set $\mathbb{k}$.

To evaluate the weight function of the NADiS, we state the following Lemma.

Lemma 3: For an ULA whose sensor positions are given as,

$$\Omega = [\mathbb{A}, \mathbb{A} + \mathbb{L}, \dots, \mathbb{A} + (\mathbb{N} - 1)\mathbb{L}]$$

the weight function corresponding to SDCA is given as,

$$w(l) = \begin{cases} \mathbb{N} - i, & l = i\mathbb{L}, \quad 0 \leq i \leq (\mathbb{N} - 1) \\ i + 1, & l = 2\mathbb{A} + i\mathbb{L}, \quad 0 \leq i \leq \mathbb{N} - 1 \\ 2\mathbb{N} - 1 - i, & l = 2\mathbb{A} + i\mathbb{L}, \quad \mathbb{N} \leq i \leq 2(\mathbb{N} - 1) \end{cases} \quad (32)$$

Proof: The location of sensors in the virtual array is given by both addition and subtraction of physical sensor positions. The weight function of the virtual array obtained from Ω is evaluated in two parts, first the weight function corresponds to the difference virtual sensors, i.e., sensors obtained by the difference of physical sensors is evaluated. Define a matrix $\Omega_d(m, n)$ as,

$$\Omega_d(m, n) = \Omega(m) - \Omega(n) \quad 1 \leq m, n \leq \mathbb{N}$$

where $\Omega(m)$ is the m th sensor of above defined ULA. The entries of matrix Ω_d enumerate the values of difference virtual array as shown in Fig. 4. It is clear from matrix Ω_d that entries along various diagonals are equal. All entries of the main diagonal are zero, all the entries below the principal diagonal are $\mathbb{L}$

	$x(A)$	$x(A+L)$	$x(A+(N-1)L)$	
$x(A)$	0	$-L$	$\cdot$	$-(N-1)L$
$x(A+L)$	L	0	$\cdot$	$-(N-2)L$
$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$
$x(A+(N-1)L)$	$(N-1)L$	$(L-2)L$	$\cdot$	0
	$\{f\}$	$\cdot$	$\cdot$	$\cdot$

(a) Difference Co-array

	$x(A)$	$x(A+L)$	$x(A+(N-1)L)$	
$x(A)$	$2A$	$2A+L$	$\cdot$	$2A+(N-1)L$
$x(A+L)$	$2A+L$	$2A+2L$	$\cdot$	$2A+N L$
$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$
$x(A+(N-1)L)$	$2A+(N-1)L$	$2A+N L$	$\cdot$	$2A+2(N-1)L$
	$\{f\}$	$\cdot$	$\cdot$	$\cdot$

(b) Sum Co-array

Fig. 4. Sum and Difference co-array of Array Ω . (Note: The number in the curly brackets $\{ \cdot \}$ denotes the frequency of corresponding entry.)

and above the main diagonal are $-L$ and so on. Based on this observation, the weight function for difference co-array $w_{\text{Diff}}(l)$ is simply given as,

$$w_{\text{Diff}}(l) = \mathbb{N} - i \quad l = iL, 0 \leq i \leq (\mathbb{N} - 1)$$

Similarly, to evaluate the sum virtual array define matrix Ω_s as,

$$\Omega_s(m, n) = \Omega(m) + \Omega(n) \quad 1 \leq m, n \leq \mathbb{N}$$

Fig. 4b shows the matrix Ω_s . Using the similar methodology the unique entries of Ω_s are calculated and weight function for sum co-array $w_{\text{Sum}}(l)$ is given as,

$$w_{\text{Sum}}(l) = \begin{cases} i + 1, & l = 2A + iL, 0 \leq i \leq \mathbb{N} - 1 \\ 2\mathbb{N} - 1 - i, & l = 2A + iL, \mathbb{N} \leq i \leq 2(\mathbb{N} - 1) \end{cases} \quad (33)$$

The weight function $w_{\text{ULA}}(l)$ corresponding to the complete SDCA of Ω is given as,

$$w_{\text{ULA}}(l) = w_{\text{Diff}}(l) \uplus w_{\text{Sum}}(l)$$

where $\uplus$ denotes the Multiset sum operation [37]. ■

The NADiS consists of the two ULAs, therefore the weight function of individual ULAs can be obtained using the above stated Lemma 3.

Corollary 1: The weight function of ULA-I ($w_{\text{ULA-I}}(l)$) of NADiS structure, having N_1 sensors with one unit inter-sensor spacing can be obtained by substituting $A = 0$, $L = 1$ and

	$x(0)$	$x(1)$	$\cdot$	$x(N_1-1)$
$x(L)$	L	$L-1$	$\cdot$	$L-(N_1-1)$
$x(L+D)$	$L+D$	$L+D-1$	$\cdot$	$L+D-(N_1-1)$
$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$
$x(L+(N_2-1)D)$	$L+(N_2-1)D$	$L+(N_2-1)D-1$	$\cdot$	$L+(N_2-1)D-N_1+1$

(a) Cross Difference Co-array

	$x(L)$	$x(L+D)$	$\cdot$	$x(L+(N_2-1)D)$
$x(0)$	L	$L+D$	$\cdot$	$L+(N_2-1)D$
$x(1)$	$L+1$	$L+D+1$	$\cdot$	$L+(N_2-1)D+1$
$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$
$x(N_1-1)$	$L+(N_1-1)$	$L+D+N_1-1$	$\cdot$	$L+(N_2-1)D+N_1-1$

(b) Cross Sum Co-array

Fig. 5. Cross Sum Co-array and Cross Difference Co-array between ULA-I and ULA-II of NADiS.

$\mathbb{N} = N_1$ in Lemma 3.

$$w_{\text{ULA-I}}(l) = \begin{cases} N_1 + 1, & l \in [0, \dots, N_1 - 1] \\ 2N_1 - l - 1, & l \in [N_1, \dots, 2N_1] \end{cases} \quad (34)$$

Corollary 2: The weight function of ULA-II ($w_{\text{ULA-II}}(l)$) of NADiS structure, having N_2 sensors with D inter-sensor spacing can be obtained by substituting $A = L$, $L = D$ and $\mathbb{N} = N_2$ in Lemma 3.

$$w_{\text{ULA-II}}(l) = \begin{cases} N_2 - i, & l = iD, \quad 0 \leq i \leq (N_2 - 1) \\ i + 1, & l = 2L + iD \quad 0 \leq i \leq N_2 - 1 \\ 2N_2 - 1 - i, & l = 2L + iD, \\ & N_2 \leq i \leq 2(N_2 - 1) \end{cases} \quad (35)$$

Further, the virtual array has the sensors corresponding to the cross sum and cross difference position of both ULAs. The weight function of these sensors is evaluated using the lemma given below.

Lemma 4: The weight function $w_{\text{cross}}(l)$ of the sensors of cross difference and cross sum between ULA-I and ULA-II is given as

$$w_{\text{cross}}(l) = \begin{cases} 1, & l \in \Omega_{\text{cd}} \\ 1, & l \in \Omega_{\text{cs}} \end{cases} \quad (36)$$

where $\Omega_{\text{cd}} = n_1 - n_2$, $n_1 \in \mathbb{S}_1, n_2 \in \mathbb{S}_2$ and $\Omega_{\text{sd}} = n_1 + n_2$, $n_1 \in \mathbb{S}_1, n_2 \in \mathbb{S}_2$.

Proof: The cross difference matrix represents the difference term between two arrays $\mathbb{S}_1$ and $\mathbb{S}_2$ which is shown in the Fig. 5a. Thus, the entries of matrix Ω_{cd} enumerate all the possible entries of cross difference virtual array. The frequency of each unique term can be calculated by the corresponding weight function. It

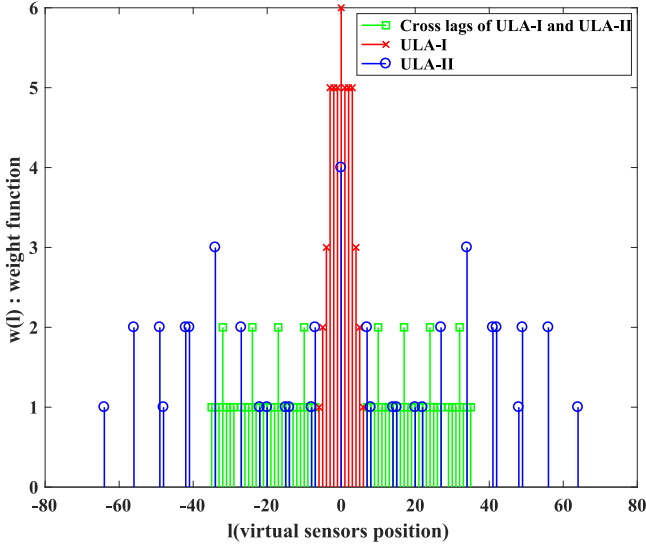


Fig. 6. Weight function of NADiS whose physical sensors located at $\{0, 1, 3, 4, 13, 22, 31\}$.

is clear from the matrix Ω_{cd} that all entries are unique. Based on this observation, the weight function of the cross difference virtual array $w_{\text{cross diff}}(l)$ is simply given as,

$$w_{\text{cross diff}}(l) = 1 \quad l \in \Omega_{cd}$$

Similarly, cross sum matrix $\Omega_{sd} = n_1 + n_2, n_1 \in \mathbb{S}_1, n_2 \in \mathbb{S}_2$ is shown in Fig. 5b. Here, all the entries are unique, so the weight function of cross sum virtual array $w_{\text{cross sum}}(l)$ is simply given as,

$$w_{\text{sum cross}}(l) = 1 \quad l \in \Omega_{cs}$$

The complete cross lag weight function is given as,

$$w_{\text{cross}}(l) = w_{\text{sum cross}}(l) \uplus w_{\text{diff cross}}(l) \quad l \in \Omega_{cs} \cup \Omega_{cd}$$

The complete weight function of NADiS structure is given as,

$$w_{\text{NADiS}}(l) = w_{\text{ULA-I}}(l) \uplus w_{\text{ULA-II}}(l) \uplus w_{\text{cross}}(l) \\ - (L + (N_2 - 1)D) \leq l \leq L + (N_2 - 1)D \quad (37)$$

The weight function of the virtual array designed in Example 2 is plotted in the Fig. 6. The weight function of ULA-I, ULA-II and cross lags between the ULA-I and ULA-II of NADiS are plotted with different color and marker to show the required contribution. It can be easily checked that the weight function given by above Lemma and numerically evaluated are same.

VI. CRAMÉR–RAO BOUND

The CRB offers a lower bound on the variance of unbiased estimates of the parameter (DoA), and it is used as a universal tool to evaluate the performance of DoA estimation algorithms [38]–[40]. The CRB for the normalized DoAs $\bar{\theta} = [\bar{\theta}_1 \dots \bar{\theta}_M]$

can be expressed as,

$$\text{CRB}_{\bar{\theta}} = \frac{1}{K} \left[\mathbf{H}_0^H (\mathbf{R}_z^T \otimes \mathbf{R}_z)^{-1} \mathbf{H}_0 \right]^{-1} \quad (38)$$

where K is the number of snapshots, $\mathbf{R}_z$ is given in (6) and $\mathbf{H}_0$ is defined in (50) of Appendix-II.

Proof: The proof is provided in Appendix II. ■

The CRB depends on the number of snapshots, sources, aperture of VULA and noise variance. The proposed CRB is numerically evaluated to study the performance of proposed structure in the subsequent section. The proposed CRB can be calculated for other arrays [25]–[27] with little modification.

VII. SIMULATION RESULTS

In this section, a series of numeric simulation results are presented to study the performance of the proposed array structure in terms of DoA estimation and compared with other existing similar array structures, i.e., Coprime Array with Sum Difference co-array (CPASD) [25] and SDCC [26]. Two array configurations are used to study the numerical results, the first array configuration has 8 physical sensors with location $S = \{0, 1, 2, 3, 4, 13, 22, 31\}$ receiving 27 uncorrelated equipowered source signals which are located uniformly between -60° to 60° with the step of 4.5° and the other configuration has 6 physical sensors array with location $S = \{0, 1, 2, 3, 10, 17\}$ receiving 17 uncorrelated equipowered source signals which are located uniformly between -60° to 60° with the step of 7.3° . The sensor locations of the CPASD [25], and the SDCC [26] with 6 physical sensors are $S_{16} = \{0, 3, 4, 6, 8, 9\}$ and $S_2 = \{0, 1, 2, 8, 12, 17\}$ respectively. For the 8 physical sensors, CPASD [25] array geometry is $S_{18} = \{0, 4, 5, 8, 10, 12, 15, 16\}$.

First, we compare the spatial spectrum of NADiS and CPASD [25] for the first array configuration. The subspace based method Spatial Smoothing based MuSiC algorithm (SS) [11]–[13] [34], Compressive Sensing (CS) [13], Direct Augmented Approach (DAA) [33], and Vandermonde Decomposition (VD) [32] are used to estimate DoAs. Afterwards, we conducted 1000 Monte-Carlo experiments to analyze the performance in terms of Mean Square Error (MSE) to estimate normalized DoA ($\bar{\theta} = \sin(\theta)/2$) as the function of input signal-to-noise ratio (SNR), number of snapshots, number of sensors, the number of sources and also with the separation between the sources.

A. Spatial Spectrum

Fig. 7 and 8 demonstrate the spatial spectra estimated using CS [13] and DAA [33] algorithms at 30dB SNR using 600 snapshots for CPASD [25] and the proposed array respectively. The results obtained from the CS are shown in Fig. 7 where grid index is 0.1° and regularization parameter $\lambda_t = 0.25$. The dashed lines in Figs. 7 and 8 are the underlying DoA and the solid lines represent the estimates of DoA. SNR is defined as

$$\text{SNR} = 10 \log \frac{(q + \bar{q} + q^* + \bar{q}^*)}{4\sigma_n^2}$$

where σ_n^2 is the variance of noise. The identifiability of the CPASD with $M + N - 1 = 8$ physical sensors is $MN +$

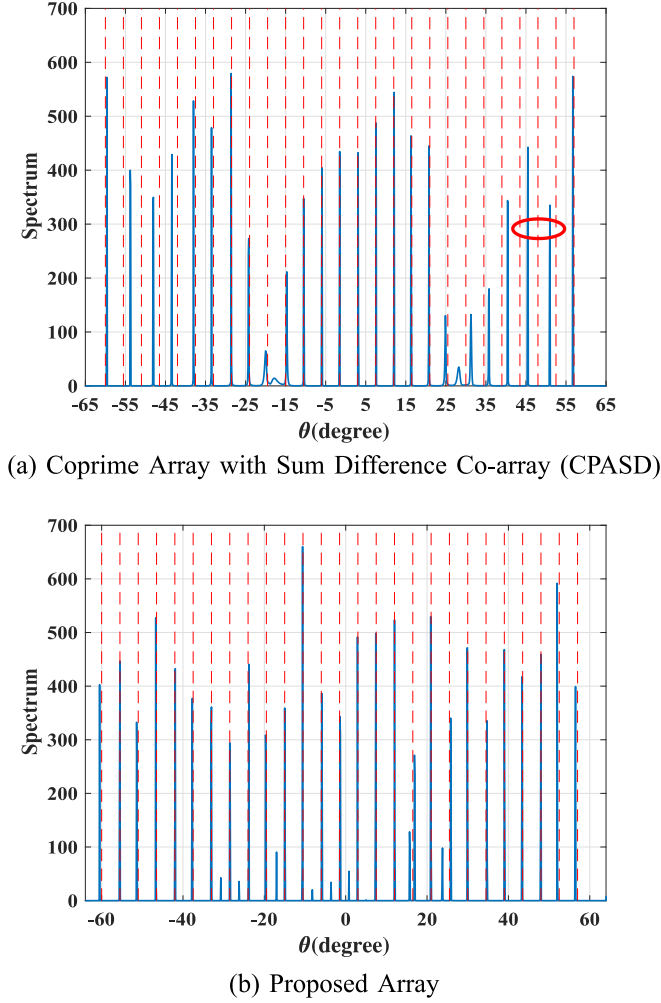


Fig. 7. Spatial Spectra estimated using Compressive sensing (CS) for both configuration with 8 sensors and $M = 27$ sources located uniformly between -60° to 60° with the step of 4.5° , 30 dB SNR and 600 snapshots.

$2M + 2N + 1 = 57$ and the proposed array identifiability with 8 sensors is $N^2 + N - 1 = 71$. we can see in Fig. 7, CPASD fail to resolve sources around 50° , i.e., it does not detect sources at 43° and 48° . However, for the same dataset, the proposed array (NADiS) correctly identifies all sources including both sources at 43° and 48° and produces a more accurate spatial spectrum. Similarly, with the DAA [33] method, NADiS efficiently identify all 27 sources compared to the CPASD [25] as clearly shown in the Fig. 8. Other algorithms also perform similarly are not included here to avoid repetition.

B. MSE Versus SNR

In the first set of simulations, we conducted Monte-Carlo experiments to analyze the performance of DoA estimation in terms of MSE as the function of input SNR. The average MSE of the estimated DoAs is given as

$$\text{MSE} = \frac{1}{I} \sum_{i=1}^I \sum_{q=1}^M (\hat{\theta}_q(i) - \theta_q)^2$$

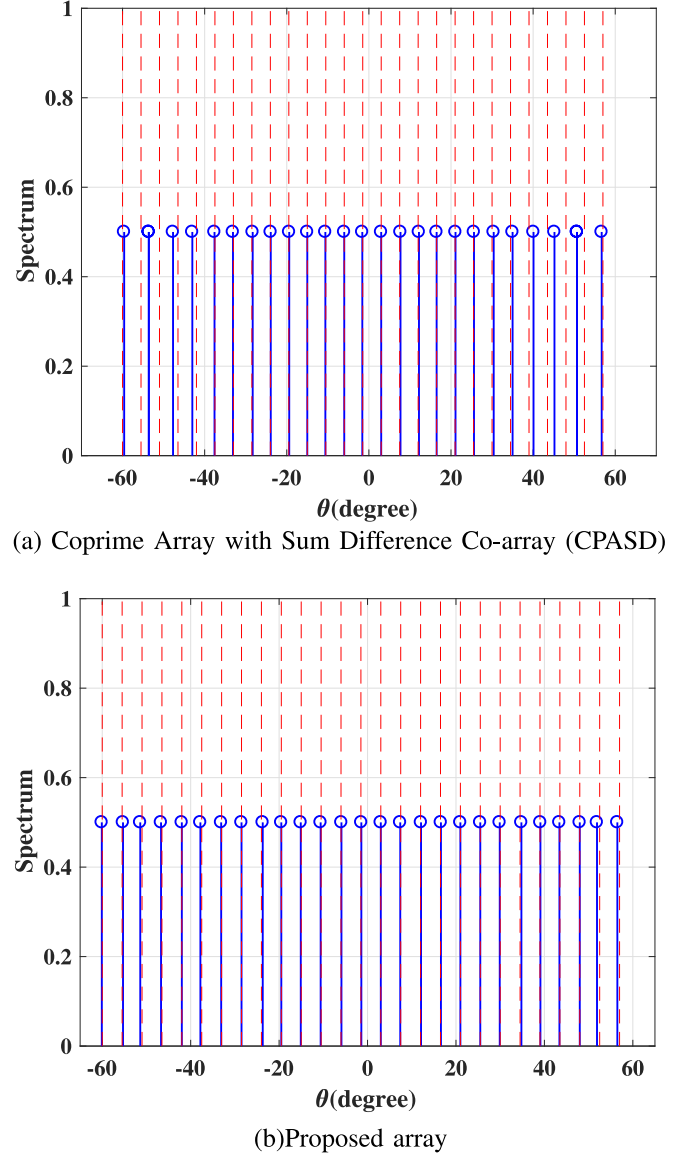


Fig. 8. Spatial Spectra estimated using Direct Augmented Approach (DAA) for both configurations with 8 sensors and $M = 27$ sources located uniformly between -60° to 60° with the step of 4.5° , 30 dB SNR and 600 snapshots.

where $\hat{\theta}_q(i)$ is the estimated DoAs for the i th Monte Carlo trial, $i = 1, \dots, I$ for q th source and θ_q is the corresponding actual DoA. The plotted MSE is averaged over all estimated DoAs of M sources. We first compare the MSE performance under different SNRs for the two array configurations as mentioned above. Fig. 9 shows the performance of the first array of 8 sensors as a function of SNR using 600 independent snapshots with various algorithms. As expected, MSE decreases as SNR increases. Though initially, at low SNR the proposed array has higher error, but at higher SNR it performs 10 times better than the CPASD structure. The accuracy of DoA estimation with SS is approximately equal to CS and others. Therefore, the study with SS will be sufficient to analyze the proposed array structure. In all other simulations, we will show only the SS results to avoid repetition as others also perform similarly.

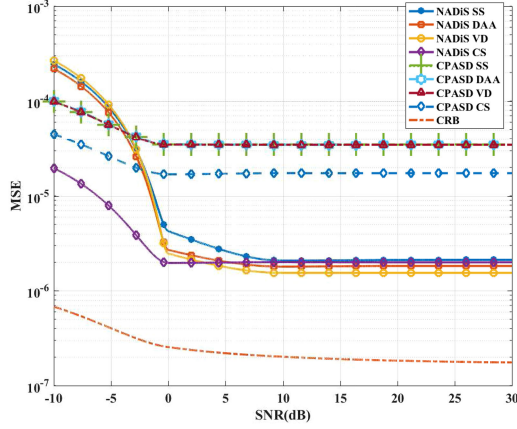


Fig. 9. Mean Square Error (MSE) vs. SNR for 1000 Monte Carlo experiments with $N = 8$ physical sensors array. The number of snapshots is $K = 600$. For $M = 27$ sources uniformly distributed in -60° to 60° with step of 4.5° .

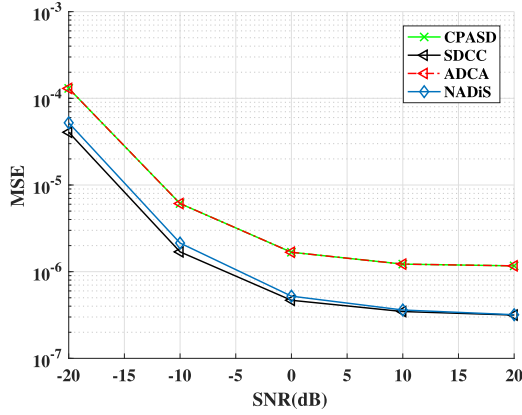


Fig. 10. The performance of the proposed CRB with SNR for ADCA, CPASD, SDCC and NADiS. The number of physical sensors array is 6. The number of snapshots is $K = 600$. For $M = 17$ sources uniformly distributed in -60° to 60° with step of 7.3° .

It is worth noting that the traditional algorithms present in literature are suggested for physical array. These algorithms are not optimal for the virtual arrays generated using covariance and pseudo covariance lags in any sense. This is clear from Fig. 9, the effect of SNR is diminishing at high SNR, i.e., no array achieves the CRB even at high SNR. This paper is focused towards the design of physical arrays so we have used only existing techniques. A better suited algorithm for these array geometries is definitely required. There exists no literature to the best of our knowledge that addresses DoA estimation for virtual arrays. It can be a topic of research on its own. But definitely, we can resolve the 35 sources using only 8 sensors and perform better than other suggested arrays.

To compare the capabilities of each array it might be worth studying their performance in terms of CRB. Fig. 10 shows the CRB as a function of SNR for different array configuration for the second case, i.e., 17 source signals uniformly distributed in -60° to 60° with step of 7.3° impinging upon the 6 sensor arrays using 600 snapshots. The CRB of NADiS and SDCC is approximately equal, this is because both designs have equal

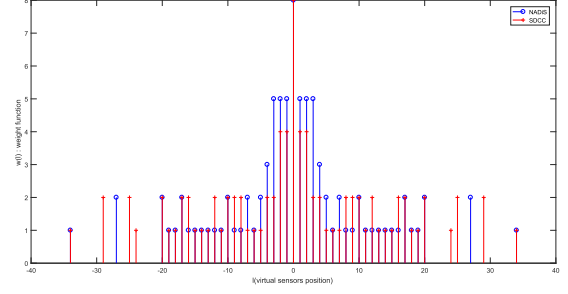


Fig. 11. Weight Function of proposed array structure for 6 physical sensors and SDCC structure.

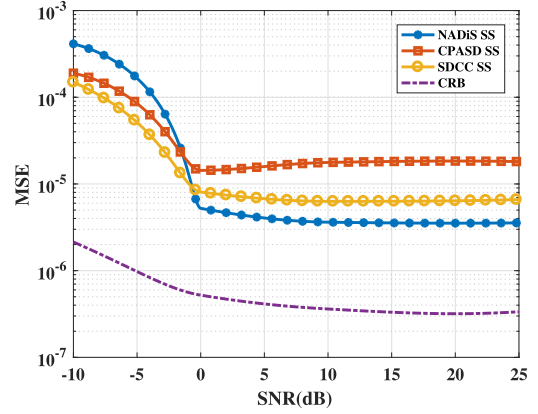


Fig. 12. Mean Square Error (MSE) vs. SNR for 1000 Monte Carlo experiments with $N = 6$ physical sensors with the location of $\{0, 1, 2, 3, 10, 17\}$. The number of snapshots is $K = 600$. For $M = 17$ sources uniformly distributed in -60° to 60° with step of 7.3° .

aperture of VULA as given in Table I. But the CRB of proposed array is less than CRB of ADCA and CPASD. It can be seen that CRB is a function of the aperture of VULA, i.e., the lower bound of performance decreases with the increasing the aperture of VULA.

When the SS is applied to the second array configuration of 6 physical sensors, we again observe that accuracy of the proposed array is better than the CPASD [25] as shown in the Fig. 12. Moreover, the NADiS achieves two times lesser error as compared to the SDCC structure [26] even with the same aperture. This is because the proposed NADiS structure is more stable than SDCC as can be inferred from Fig. 11 which it is showing the weight function of both the arrays. The NADiS used most of the lags in VULA whereas SDCC used less number of lags in VULA.

To study the efficacy of NADiS array structure in a more general setting, we compared 6 sensors NULA, ULA having $\lambda/2$ inter-sensor spacing and the virtual array given in (21). Fig. 13 shows the performance as a function of SNR for 5 sources at $\{-60, -33, -6, 21, 48\}$ using 400 snapshots. Here, NULA and ULA have only 6 sensors so they can resolve only 5 sources. It is clear from the Fig. 13 that the proposed virtual array has lower threshold than ULA and also attains lower MSE even at higher SNR, it may be because of the lower aperture of ULA. NULA has higher aperture so performs better but the

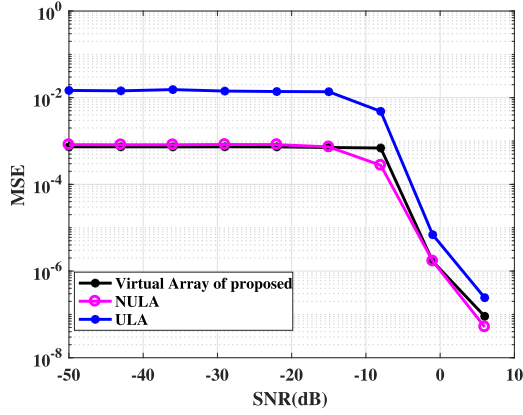


Fig. 13. Mean Square Error (MSE) vs. SNR for 1000 Monte Carlo experiments with $N = 6$ physical sensors with the location of $\{0, 1, 2, 3, 10, 17\}$. The number of snapshots is $K = 400$. For $M = 5$ sources located at $\{-60, -33, -6, 21, 48\}$.

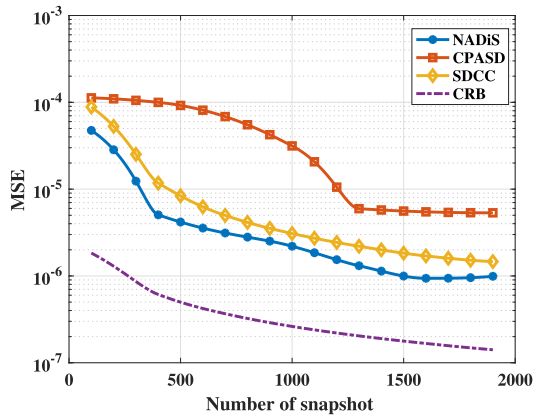


Fig. 14. Mean Square Error (MSE) vs. snapshot for 1000 Monte Carlo experiments with $N = 6$ physical sensors with the location of $\{0, 1, 2, 3, 10, 17\}$. For $M = 17$ sources uniformly distributed in -60° to 60° with step of 7.3° . Here SNR = 30 dB.

performance of suggested virtual array is comparable to NULA. Therefore, one can use the NADiS structure based virtual array for even fewer sources without compromising performance.

C. MSE Versus Snapshots

Next, we fix the SNR at 30 dB and study the performance w.r.t. the number of snapshots for array configuration with 6 physical sensors as mentioned above. Fig. 14 shows the performance of all array structures with the number of snapshots. It can be seen from the Fig. 14 that the NADiS with SS shows significantly better performance as compared to the other two structures [25]–[26].

D. MSE Versus Number of Sources

We further study the estimation accuracy of NADiS with number of sources. We analyze both array configurations at 30 dB SNR and using 800 snapshots. Fig. 15 and Fig. 16 compare the performance of the arrays with 6 physical sensors and 8 physical sensors respectively and clearly show that NADiS

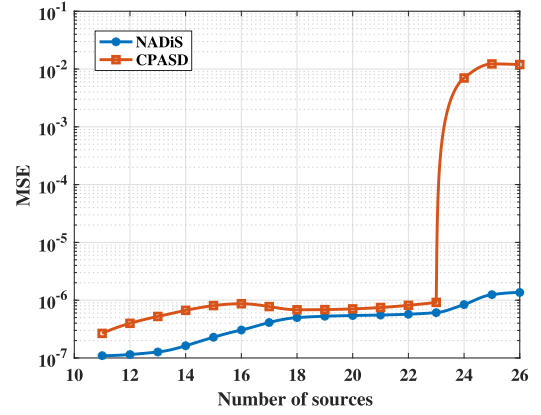


Fig. 15. Mean Square Error (MSE) vs. sources for 1000 Monte Carlo experiments with $N = 8$ physical sensors with the location of $\{0, 1, 2, 3, 4, 13, 22, 31\}$. The number of snapshots is $K = 800$ and SNR = 30 dB. For $M = 17$ sources uniformly distributed in -60° to 60° with step of 4.5° .

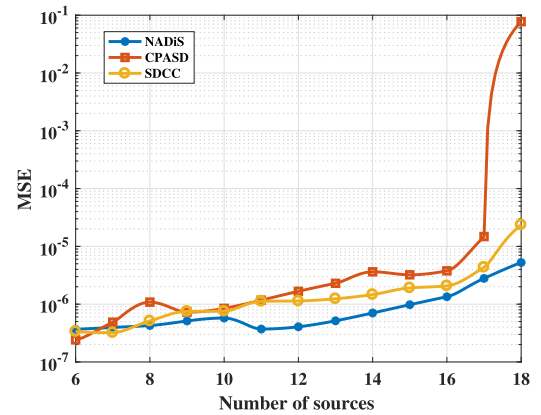


Fig. 16. Mean Square Error (MSE) vs. sources for 1000 Monte Carlo experiments with $N = 6$ physical sensors with the location of $\{0, 1, 2, 3, 10, 17\}$. The number of snapshots is $K = 800$ and SNR = 30 dB. For $M = 17$ sources uniformly distributed in -60° to 60° with step of 7.3° .

offers high performance gain for large number of sources. It is clearly seen from Fig. 15 and Fig. 16 that NADiS performs better than CPASD in all scenarios. Moreover, CPASD ceases to perform for more than 21 sources in 8 array configuration and 17 sources in 6 array configuration whereas the proposed structure with 8 physical sensors can even resolve 26 sources and 6 physical sensor array can resolve 18 sources with high accuracy.

E. MSE Versus Sensors

In this section, we investigate how the number of sensors affects the MSE. In this experiment, we consider $M = 27$ sources, SNR = 20 dB and snapshots $K = 600$. Fig. 17 summarizes the results, it is observed that to achieve MSE to the order of 10^{-7} , NADiS are required only 13 sensors whereas 16 sensors for CPASD. Another interesting observation is that CPASD has high threshold as compared to NADiS.

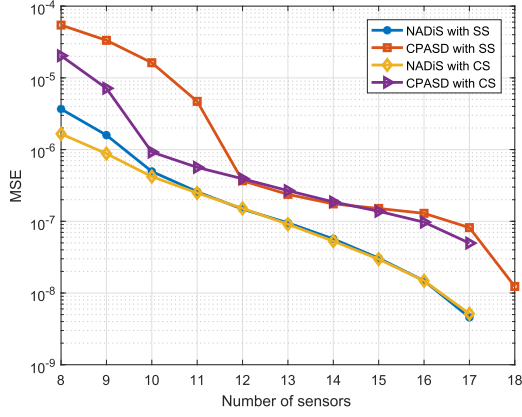


Fig. 17. Mean Square Error (MSE) vs. Number of sensors for 1000 Monte Carlo experiments with $M = 27$ sources uniformly distributed in -60° to 60° with step of 4.5° . The number of snapshots is $K = 600$ and SNR = 20 dB.

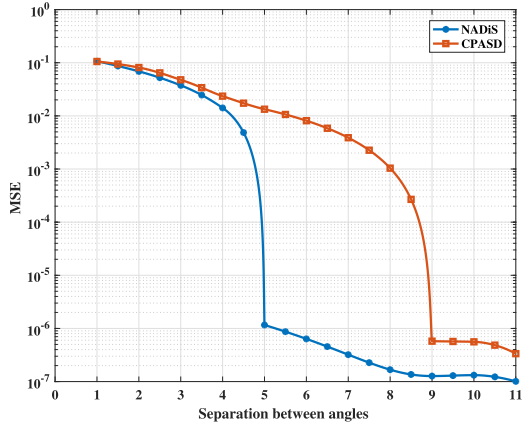


Fig. 18. Mean Square Error (MSE) vs. Separation between Sources for 1000 Monte Carlo experiments with $N = 8$ physical sensors with the location of $\{0, 1, 2, 3, 4, 13, 22, 31\}$. The number of snapshots is $K = 600$ and SNR = 30 dB. For $M = 11$ sources uniformly distributed in -60° to 60° with step of 12° .

F. MSE Versus Separation Between Sources

Fig. 18 shows MSE versus separation between 11 sources using 600 snapshots. We consider the 8 physical sensor array structure. It can be seen from Fig. 18 that the estimation performance of the proposed array increases as the separation between the sources increases. With NADiS MSE in the order of 10^{-5} is achieved at separation 5° while CAPSD achieves this MSE at 9° separation. Therefore, we can say that the suggested array structure is much more effective and the estimation is much better as compared to earlier suggested structures.

VIII. CONCLUSION

We have proposed a novel array structure named as NADiS, which provides the high aperture VULA and enhances the DoA estimation performance. We derived the analytic expression of the aperture of the virtual array and the optimum length of both ULAs. We analyzed the stability of the virtual array by using the weight function and obtained the close form expression for the weight function of the proposed array. Also, to evaluate the performance of the proposed array, the universal tool CRB, is

derived. Exhaustive numeric simulation studies demonstrate the efficiency of the proposed array to estimate DoAs of noncircular signals in different scenarios.

APPENDIX I DOA ESTIMATING ALGORITHM

A. Spatial Smoothing Based MuSiC Algorithm

The vector $\mathbf{r}_z$ represents a single snapshot of received data from the longer aperture virtual array, therefore, the covariance matrix evaluated using single snapshot data, is an ill conditioned covariance matrix. To build a full rank covariance matrix, spatial smoothing based [11]–[13] technique is used so that traditional subspace methods can be applied. The $\mathbf{r}_z$ is divided into T overlapping subarrays with Q sensors and the t -th subarray sensors are located from $(-t + 1)$ to $(-t + Q)$. The received signal vector for the t th subarray is denoted as

$$\mathbf{w}_{1t} = \mathbf{B}_{1t} \mathbf{p} \quad t = 1, 2, \dots, T$$

where $\mathbf{B}_{1t} \in \mathbb{C}^{Q \times L}$ denotes the steering matrix of t th subarray for L sources. A full rank spatially smoothed positive definite matrix $\mathbf{R}_{ss}$ can be obtained by averaging the all full covariance matrix $\mathbf{R}_t$, which is constructed using $\mathbf{w}_{1t}$ as,

$$\mathbf{R}_t = \mathbf{E}[\mathbf{w}_{1t} \mathbf{w}_{1t}^H]$$

The spatially smoothed matrix,

$$\mathbf{R}_{ss} = \frac{1}{T} \sum_{t=1}^T \mathbf{w}_{1t} \mathbf{w}_{1t}^H$$

Here, we assume $Q > L$ therefore this covariance matrix will be full rank, and can identify up to $Q - 1$ sources. Finally, we can apply subspace based algorithms like MuSiC, Root MuSiC, etc. on the spatially smoothed covariance matrix $\mathbf{R}_{ss}$ to obtain the information of DoA.

APPENDIX II CRAMÉR–RAO BOUND

The CRB is defined as the inverse of the Fisher Information Matrix (FIM) $[\mathbb{I}(\bar{\theta})]_{i,j}$ [38]. For the given array structure, we evaluate the CRB for normalized DoAs ($\bar{\theta}$) using the covariance matrix $\mathbf{R}_z$ as follows,

$$[\mathbb{I}(\bar{\theta})]_{i,j} = K \text{tr} \left(\mathbf{R}_z^{-1} \frac{\partial \mathbf{R}_z}{\partial \bar{\theta}_i} \mathbf{R}_z^{-1} \frac{\partial \mathbf{R}_z}{\partial \bar{\theta}_j} \right) \quad (39)$$

where $\bar{\theta} = (d/\lambda) \sin(\theta)$ is the unknown parameter to be estimated using K snapshots and $\text{tr}(\cdot)$ denotes the trace operator. Using the identity, $\text{tr}(\mathbf{A}\mathbf{B}\mathbf{C}\mathbf{D}) = \text{vec}(\mathbf{B}^H)(\mathbf{A}^T \otimes \mathbf{C})\text{vec}(\mathbf{D})$ [38] and (7), the above equation (39) becomes,

$$[\mathbb{I}(\bar{\theta})]_{i,j} = \left[(\mathbf{R}_z^T \otimes \mathbf{R}_z)^{-1/2} \frac{\partial \mathbf{r}_z}{\partial \bar{\theta}_i} \right]^H \left[(\mathbf{R}_z^T \otimes \mathbf{R}_z)^{-1/2} \frac{\partial \mathbf{r}_z}{\partial \bar{\theta}_j} \right] \quad (40)$$

The FIM $\mathbb{I}$ can be written as follows,

$$\mathbb{I}(\bar{\theta}) = \mathbf{V}^H \mathbf{V} \quad (41)$$

where

$$\mathbf{V} = (\mathbf{R}_z^T \otimes \mathbf{R}_z)^{-1/2} \left[\frac{\partial \mathbf{r}_z}{\partial \theta_1} \dots \frac{\partial \mathbf{r}_z}{\partial \theta_M} \right] \quad (42)$$

From (11), we get $\mathbf{r}_z$

$$\mathbf{r}_z = \mathbf{J} \begin{bmatrix} \sum_{i=1}^M (\mathbf{a}^*(\bar{\theta}_i) \otimes \mathbf{a}(\bar{\theta}_i)) g_i \\ \sum_{i=1}^M (\mathbf{a}^*(\bar{\theta}_i) \otimes \mathbf{a}^*(\bar{\theta}_i)) \tilde{g}_i^* \\ \sum_{i=1}^M (\mathbf{a}(\bar{\theta}_i) \otimes \mathbf{a}(\bar{\theta}_i)) \tilde{g}_i \\ \sum_{i=1}^M (\mathbf{a}(\bar{\theta}_i) \otimes \mathbf{a}^*(\bar{\theta}_i)) g_i^* \end{bmatrix} + \mathbf{J} \begin{bmatrix} \sigma_n^2 \mathbf{1}_n \\ \mathbf{0} \\ \mathbf{0} \\ \sigma_n^2 \mathbf{1}_n \end{bmatrix} \quad (43)$$

Here, $\mathbf{a}^*(\bar{\theta}_i) \otimes \mathbf{a}(\bar{\theta}_i)$ corresponds to the positive difference co-array as given in (17). Let $\mathbf{c}_D^+(\bar{\theta}_i)$ be the steering vector corresponding to positive difference co-array ($\mathbf{S}_D^+$). Using $\mathbf{c}_D^+(\bar{\theta}_i)$, we can write

$$\mathbf{a}^*(\bar{\theta}_i) \otimes \mathbf{a}(\bar{\theta}_i) = \mathbb{K}_1 \mathbf{c}_D^+(\bar{\theta}_i) \quad (44)$$

where $\mathbb{K}_1$ is the mapping matrix for difference co-array, has size $N^2 \times |\mathbf{S}_D^+|$, defined as,

$$\langle \mathbb{K}_1 \rangle_{:,q} = \text{vec}(\mathbb{I}_1(q)) \quad q \in \mathbf{S}_D^+ \quad (45)$$

where $\mathbb{I}_1(q)$ is $N \times N$ matrix, defined as,

$$\langle \mathbb{I}_1(q) \rangle_{n_1, n_2} = \begin{cases} 1, & n_1 - n_2 = q \\ 0, & \text{otherwise} \end{cases}, \quad n_1, n_2 \in \mathbb{S}_{pa} \quad (46)$$

Similarly, other terms of (44),

$$\begin{aligned} \mathbf{a}(\bar{\theta}_i) \otimes \mathbf{a}(\bar{\theta}_i) &= \mathbb{K}_2 \mathbf{c}_S^+(\bar{\theta}_i) \\ \mathbf{a}^*(\bar{\theta}_i) \otimes \mathbf{a}^*(\bar{\theta}_i) &= \mathbb{K}_3 \mathbf{c}_S^-(\bar{\theta}_i) \\ \mathbf{a}(\bar{\theta}_i) \otimes \mathbf{a}^*(\bar{\theta}_i) &= \mathbb{K}_4 \mathbf{c}_D^-(\bar{\theta}_i) \end{aligned} \quad (47)$$

where $\mathbb{K}_2, \mathbb{K}_3$ and $\mathbb{K}_4$ is same as (45) for other co-arrays. After substituting (44) and (47) in (11), (42) becomes,

$$\mathbf{V} = (\mathbf{R}_z^T \otimes \mathbf{R}_z)^{-1/2} \mathbf{J} \begin{bmatrix} \mathbb{K}_1 \left[\frac{\partial \mathbf{c}_D^+(\bar{\theta}_1)}{\partial \theta_1} g_1 \dots \frac{\partial \mathbf{c}_D^+(\bar{\theta}_M)}{\partial \theta_M} g_M \right] \\ \mathbb{K}_3 \left[\frac{\partial \mathbf{c}_S^-(\bar{\theta}_1)}{\partial \theta_1} \tilde{g}_1^* \dots \frac{\partial \mathbf{c}_S^-(\bar{\theta}_M)}{\partial \theta_M} \tilde{g}_M^* \right] \\ \mathbb{K}_2 \left[\frac{\partial \mathbf{c}_S^+(\bar{\theta}_1)}{\partial \theta_1} \tilde{g}_1 \dots \frac{\partial \mathbf{c}_S^+(\bar{\theta}_M)}{\partial \theta_M} \tilde{g}_M \right] \\ \mathbb{K}_4 \left[\frac{\partial \mathbf{c}_D^-(\bar{\theta}_1)}{\partial \theta_1} g_1^* \dots \frac{\partial \mathbf{c}_D^-(\bar{\theta}_M)}{\partial \theta_M} g_M^* \right] \end{bmatrix} \quad (48)$$

After differentiation, we obtain,

$$\mathbf{V} = (\mathbf{R}_z^T \otimes \mathbf{R}_z)^{-1/2} \mathbf{J} \begin{bmatrix} \mathbb{K}_1 j2\pi \text{diag}(\mathbf{S}_D^+) \mathbf{C}_D^+ \mathbf{G} \\ \mathbb{K}_3 j2\pi \text{diag}(\mathbf{S}_S^-) \mathbf{C}_S^- \tilde{\mathbf{G}}^* \\ \mathbb{K}_2 j2\pi \text{diag}(\mathbf{S}_S^+) \mathbf{C}_S^+ \tilde{\mathbf{G}} \\ \mathbb{K}_4 j2\pi \text{diag}(\mathbf{S}_D^-) \mathbf{C}_D^- \mathbf{G}^* \end{bmatrix} \quad (49)$$

where $\mathbf{G} = \text{diag}(g_1 \dots g_M)$, $\tilde{\mathbf{G}} = \text{diag}(\tilde{g}_1 \dots \tilde{g}_M)$, and $\text{diag}(\mathbf{S}_D^+)$ denotes diagonal matrix with $\mathbf{S}_D^+$ as its diagonal. Similarly, other diagonal matrices have $\mathbf{S}_S^+$, $\mathbf{S}_S^-$ and $\mathbf{S}_D^-$ as their diagonals is represented the manifold matrix corresponding to positive difference co-array $\mathbf{S}_D^+$ by $\mathbf{C}_D^+$. The other $\mathbf{C}_S^+$, $\mathbf{C}_S^-$, $\mathbf{C}_D^-$ represent manifold matrix corresponding to positive sum co-array $\mathbf{S}_S^+$,

negative sum co-array $\mathbf{S}_S^-$, negative difference co-array $\mathbf{S}_D^-$. FIM matrix Equation (41) becomes,

$$\begin{aligned} \mathbb{I}(\bar{\theta}) &= K[(\mathbf{R}_z^T \otimes \mathbf{R}_z)^{-1/2} \mathbf{H}_0]^H [(\mathbf{R}_z^T \otimes \mathbf{R}_z)^{-1/2} \mathbf{H}_0] \\ &= K \mathbf{H}_0^H (\mathbf{R}_z^T \otimes \mathbf{R}_z)^{-1} \mathbf{H}_0 \end{aligned} \quad (50)$$

where

$$\mathbf{H}_0 = \mathbf{J} \begin{bmatrix} \mathbb{K}_1 j2\pi \text{diag}(\mathbf{S}_D^+) \mathbf{C}_D^+ \mathbf{G} \\ \mathbb{K}_3 j2\pi \text{diag}(\mathbf{S}_S^-) \mathbf{C}_S^- \tilde{\mathbf{G}}^* \\ \mathbb{K}_2 j2\pi \text{diag}(\mathbf{S}_S^+) \mathbf{C}_S^+ \tilde{\mathbf{G}} \\ \mathbb{K}_4 j2\pi \text{diag}(\mathbf{S}_D^-) \mathbf{C}_D^- \mathbf{G}^* \end{bmatrix}$$

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